IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE/EIE PART II: MEng, BEng and ACGI

COMMUNICATION SYSTEMS

Wednesday, 3 June 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s): J.A. Barria

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EXAM QUESTIONS

Information for Students

Fourier Transform Pairs

Pair Number	$x(t)$	$X(f)$
1.	$\Pi\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc} \tau f$
2.	$2W \operatorname{sinc} 2Wt$	$\Pi\left(\frac{f}{2W}\right)$
3.	$\Lambda\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc}^2 \tau f$
4.	$\exp(-\alpha t)u(t),\ \alpha > 0$	$\frac{1}{\alpha + j2\pi f}$
5.	$t \exp(-\alpha t)u(t),\ \alpha > 0$	$\frac{1}{(\alpha + j2\pi f)^2}$
6.	$\exp(-\alpha\|t\|),\ \alpha > 0$	$\frac{2\alpha}{\alpha^2 + (2\pi f)^2}$
7.	$e^{-\pi(t/\tau)^2}$	$\tau e^{-\pi(f\tau)^2}$
8.	$\delta(t)$	1
9.	1	$\delta(f)$
10.	$\delta(t - t_0)$	$\exp(-j2\pi f t_0)$
11.	$\exp(j2\pi f_0 t)$	$\delta(f - f_0)$
12.	$\cos 2\pi f_0 t$	$\frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0)$
13.	$\sin 2\pi f_0 t$	$\frac{1}{2j}\delta(f - f_0) - \frac{1}{2j}\delta(f + f_0)$
14.	$u(t)$	$(j2\pi f)^{-1} + \frac{1}{2}\delta(f)$
15.	$\operatorname{sgn} t$	$(j\pi f)^{-1}$
16.	$\frac{1}{\pi t}$	$-j \operatorname{sgn}(f)$
17.	$\hat{x}(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(\lambda)}{t - \lambda}\, d\lambda$	$-j \operatorname{sgn}(f)X(f)$
18.	$\sum_{m=-\infty}^{\infty} \delta(t - mT_s)$	$f_s \sum_{m=-\infty}^{\infty} \delta(f - mf_s),$ $f_s = T_s^{-1}$

Useful Relations and Formulas

$$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\tan(\frac{\pi}{2} - x) = \cot x$$

$$\tan 2x = \frac{2\tan x}{1-\tan^2 x} = \frac{2\cot x}{\cot^2 x - 1} = \frac{2}{\cot x - \tan x}$$

$$\tan\frac{x}{2} = \pm\sqrt{\frac{1-\cos x}{1+\cos x}} = \frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$$

$$\sin x \cos y = \frac{1}{2}[\sin(x+y) + \sin(x-y)]$$

$$\cos x \sin y = \frac{1}{2}[\sin(x+y) - \sin(x-y)]$$

$$\sin x \sin y = \frac{1}{2}[\cos(x-y) - \cos(x+y)]$$

$$\cos x \cos y = \frac{1}{2}[\cos(x+y) + \cos(x-y)]$$

$$\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2}$$

$$\sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$$

$$\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2}$$

$$\cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}$$

Differentiation Rule of Leibnitz

Let $F(z) = \int_{a(z)}^{b(z)} f(x,z)dx$. Then we have

$$\frac{dF(z)}{dz} = \frac{db(z)}{dz}f(b(z),z) - \frac{da(z)}{dz}f(a(z),z) + \int_{a(z)}^{b(z)} \frac{\partial f(x,z)}{\partial z}dx$$

Hilbert Transform

Hilbert transform of a signal $g(t)$ is given by

$$\hat{g}(t) = \frac{1}{\pi}\int_{-\infty}^{+\infty} \frac{g(\tau)}{t-\tau}d\tau$$

Fourier Transform Theorems[a]

Name of Theorem		
1. Superposition (a_1 and a_2 arbitrary constants)	$a_1x_1(t) + a_2x_2(t)$	$a_1X_1(f) + a_2X_2(f)$
2. Time delay	$x(t - t_0)$	$X(f)e^{-j2\pi ft_0}$
3a. Scale change	$x(at)$	$\lvert a\rvert^{-1}X\left(\frac{f}{a}\right)$
b. Time reversal	$x(-t)$	$X(-f) = X*(f)$
4. Duality	$X(t)$	$x(-f)$
5a. Frequency translation	$x(t)e^{j\omega_0 t}$	$X(f - f_0)$
b. Modulation	$x(t)\cos\omega_0 t$	$\frac{1}{2}X(f - f_0) + \frac{1}{2}X(f + f_0)$
6. Differentiation	$\frac{d^n x(t)}{dt^n}$	$(j2\pi f)^n X(f)$
7. Integration	$\int_{-\infty}^{t} x(t')\,dt'$	$(j2\pi f)^{-1}X(f) + \frac{1}{2}X(0)\delta(f)$
8. Convolution	$\int_{-\infty}^{\infty} x_1(t - t')x_2(t')\,dt'$	
		$X_1(f)X_2(f)$
	$= \int_{-\infty}^{\infty} x_1(t')x_2(t - t')\,dt'$	
9. Multiplication	$x_1(t)x_2(t)$	$\int_{-\infty}^{\infty} X_1(f - f')X_2(f')\,df'$
		$= \int_{-\infty}^{\infty} X_1(f')X_2(f - f')\,df'$

Q Function

$$Q(x) = \frac{1}{\sqrt{2\pi}}\int_x^{\infty} e^{-t^2/2}dt$$

Some reference values of the Q function are given below.

x	0	0.5	1.0	1.5	2.0	2.5	3
Q(x)	0.5	0.3085	0.1587	0.0668	0.02275	0.00621	$1.35 \cdot 10^{-3}$

Joint Gaussian density

The joint probability density function (pdf) of two correlated Gaussian random variables X and Y is given by

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]}.$$

where $\mu_X = E[X]$, $\mu_Y = E[Y]$ are the mean values, σ_X and σ_Y are the standard deviation of X and Y, respectively, and ρ is the correlation coefficient defined as $\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}$.

1. a) i) Can a random process be second-order (strict sense) stationary, but not first-order stationary? [3]

ii) What is the advantage of single sideband (SSB) modulation with respect to double sideband (DSB) modulation? [2]

iii) Let $X(t)$ be the Hilbert transform of a Gaussian process. Is $X(t)$ also Gaussian? [2]

iv) Random variable X is defined on alphabet $\mathscr{X} = \{1, \ldots, n\}$, and its entropy is denoted by $H(X)$. What is the maximum value $H(X)$ can take? [2]

b) For random processes $X(t)$ and $Y(t)$ their *cross-correlation function* is defined as $R_{XY}(t_1,t_2) = E[X(t_1)Y(t_2)]$, while their cross-covariance function is defined as $C_{XY}(t_1,t_2) = E[(X(t_1) - \mu_X(t_1))(Y(t_2) - \mu_Y(t_2))]$, where $\mu_X(t) = E[X(t)]$.

$X(t)$ and $Y(t)$ are *jointly wide sense stationary (WSS)*, if they are both WSS and their crosscorrelation function $R_{XY}(t_1,t_2)$ depends only on the time difference $\tau = t_1 - t_2$, and can be denoted as $R_{XY}(\tau)$.

i) Show that $C_{XY}(t_1,t_2) = R_{XY}(t_1,t_2) - \mu_X(t_1)\mu_Y(t_2)$. [2]

ii) Let $Y(t) = X(t) + N(t)$, where $N(t)$ is a noise process independent of $X(t)$. Find the cross-covariance of $X(t)$ and $Y(t)$ in terms of the autocorrelation of $X(t)$ and $\mu_X(t)$. [3]

iii) Let $X(t)$ be a random process. Assuming the derivative is well-defined at every t, we set $Y(t) = \frac{d}{dt}X(t)$. Show that $R_{XY}(t_1,t_2) = \frac{\partial}{\partial t_2}R_X(t_1,t_2)$. [3]

iv) For jointly WSS processes $X(t)$ and $Y(t)$, show that $R_{XY}(\tau) \leq \frac{1}{2}[R_X(0) + R_Y(0)]$. [4]

c) An analog signal with bandwidth 3 KHz is sampled and then quantized with a uniform scalar quantizer. For each quantized sample, the quantization error should not exceed the 0.1% of the peak-to-peak range of the analog signal.

i) What is the minimum required sampling rate? [2]

ii) What is the minimum number of bits per sample? [3]

iii) What is the corresponding bit rate? [2]

iv) If these bits are to be transmitted using 16-ary pulse amplitude modulation (PAM), what is the required symbol duration? [3]

d) Consider baseband transmission of a signal $m(t)$. The signal is corrupted by additive zero-mean white Gaussian noise $n(t)$ with power spectral density $S_n(f) = 2,\ \forall f$.

Message $m(t)$ is a zero-mean wide sense stationary process with power spectral density given as follows

$$S_m(f) = \begin{cases} 9 - |f|, & |f| \leq 5 \\ 4, & 5 \leq |f| \leq 9 \\ 0, & \text{otherwise} \end{cases}$$

The receiver applies the following lowpass filter $H(f)$ to the received signal:

$$H(f) = \begin{cases} 2, & -5 \leq f \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

Let $y(t)$ denote the output of the filter.

i) What is the power of noise $n(t)$? [2]

ii) What is the signal to noise ratio (SNR) of $y(t)$? [4]

iii) What is the SNR if the lowpass filter cut-off frequencies are increased to $[-9, 9]$? Comment about the relative quality of the received signal $y(t)$ with respect to the previous case. [3]

2. a) Let $X(t)$ be a zero-mean wide sense stationary (WSS) Gaussian process with autocorrelation function $R(\tau) = e^{-|\tau|}\cos(2\pi\tau)$.

i) Find the joint probability density function (pdf) of $X(\frac{1}{6})$ and $X(\frac{2}{6})$. [6]

ii) Find the joint pdf of $X(\pi)$ and $X\left(\pi + \frac{1}{6}\right)$. [3]

iii) What is the power spectral density (PSD) of $X(t)$? [5]

b) A binary message source generates bit "0" with probability p_0 and "1" with probability p_1. These bits are transmitted over a binary digital communication system. Bit "0" is transmitted with a pulse of amplitude -1, and bit "1" is transmitted with a pulse of amplitude 1. The noise in the channel is zero-mean additive white Gaussian with variance 1. The receiver uses a matched filter followed by threshold detection.

Let P_{e0} denote the probability of error when bit "0" is transmitted, while P_{e1} denote the probability of error when bit "1" is transmitted. The overall probability of error is given by $P_e = p_0 \cdot P_{e0} + p_1 \cdot P_{e1}$.

i) Find p_0 if the optimal threshold that minimizes P_e is given by $T = \ln 2$. [6]

ii) Plot the trade-off between P_{e0} and P_{e1} approximately as the threshold T goes from $T = -\infty$ to $T = +\infty$. Indicate the point corresponding to $T = 0$ on the plot. [4]

iii) If $p_0 = p_1 = 0.5$, what is the minimum probability of error that can be achieved while guaranteeing that $P_{e0} \leq 0.0668$? [6]

3. a) Assume that a memoryless source outputs 5 symbols with probabilities $1/2, 1/4, 1/8, 1/16, 1/16$.

i) Design a Huffman code for compressing this source output. Show clearly the code construction process, and the codewords corresponding to each source output. [4]

ii) What is the average codeword length of this code? [2]

iii) What is the entropy of this source? [3]

iv) Can you improve the compression efficiency of this source (bits per source symbol) by compressing multiple source outcomes jointly? [3]

b) Let X and Y be jointly distributed random variables defined over sets $\mathscr{X}$ and $\mathscr{Y}$, respectively. The mutual information between X and Y is defined as

$$I(X;Y) = \sum_{x\in\mathscr{X}} \sum_{y\in\mathscr{Y}} p(x,y) \log \frac{p(x,y)}{p(x)p(y)},$$

while the conditional entropy of X given Y is defined as

$$H(X|Y) = \sum_{x\in\mathscr{X}} \sum_{y\in\mathscr{Y}} p(x,y) \log \frac{1}{p(x|y)}.$$

i) Prove that $I(X;Y) = H(X) - H(X|Y)$. [5]

ii) Prove that $H(X,Y) = H(X) + H(Y|X)$. [5]

iii) Find $I(X;Y)$ for the joint distribution given below: [8]

		Y		
P(X,Y)		a	b	c
X	1	1/16	3/16	1/16
	2	1/4	1/16	0
	3	0	1/8	1/4